

Carges and Fields

Building Models to Describe our World
Induction, Coulomb and The Electric Field



Outline

Induction

- Charging a neutral electroscope
- Dirod generator

Coulomb's Law

- Coulomb's discovery
- Recall: Newton's gravity
- Coulomb's Law
- Gravity and the electric force
- Inverse square laws and waves

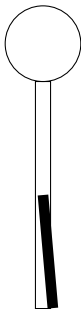
The Electric Field

- Derivation
- Field lines
- Superposition
- Electric field from a half-ring of charge
- Electric field from an infinite line charge
- 2D infinite plane
- 2D infinite plane, method 2
- Electric dipoles

Induction

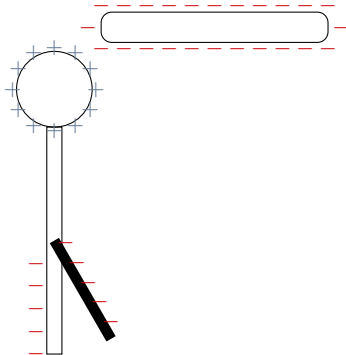
Building Models to Describe our World
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Charging by induction



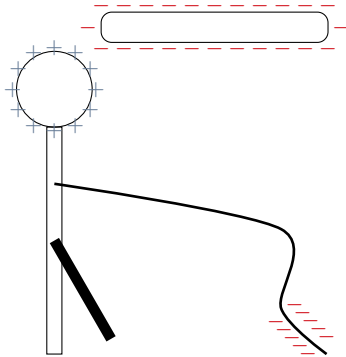
Charging an electroscope by induction. Electroscope is neutral.

Charging by induction



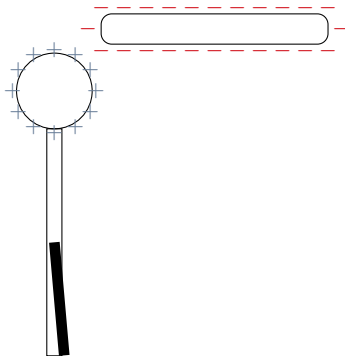
Charging an electroscope by induction. Electroscope is neutral but charges separate.

Charging by induction



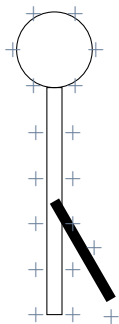
Charging an electroscope by induction. Charges opposite of those on the rod can escape through the ground wire.

Charging by induction



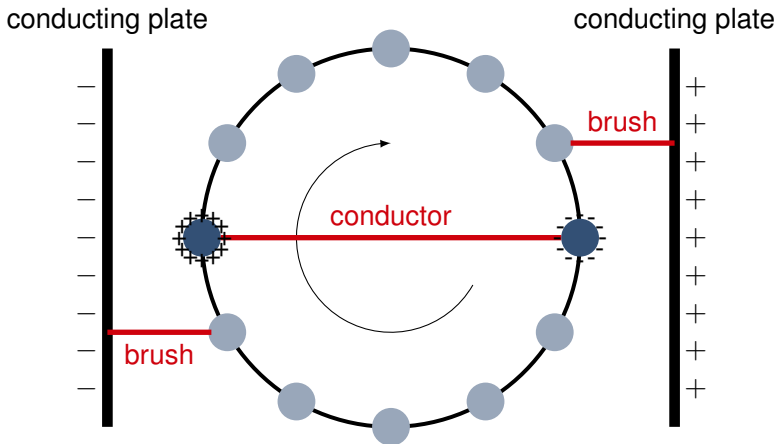
Charging an electroscope by induction. The ground wire is disconnected, positive charges stay on electroscope.

Charging by induction



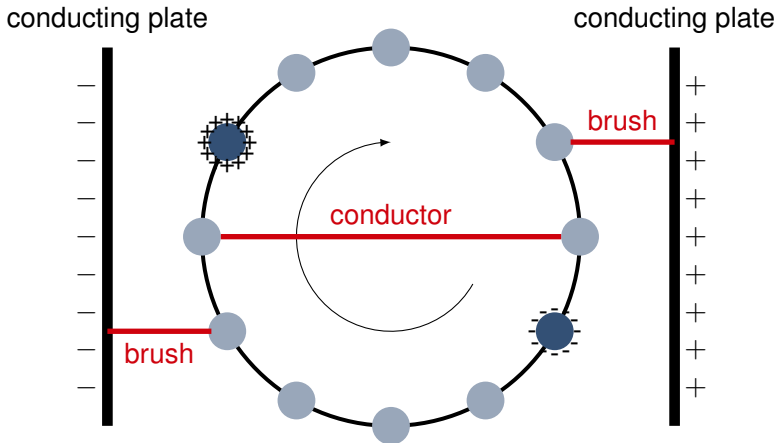
Charging an electroscope by induction. Charges redistribute along the electroscope, leaving it charged.

The dirod generator



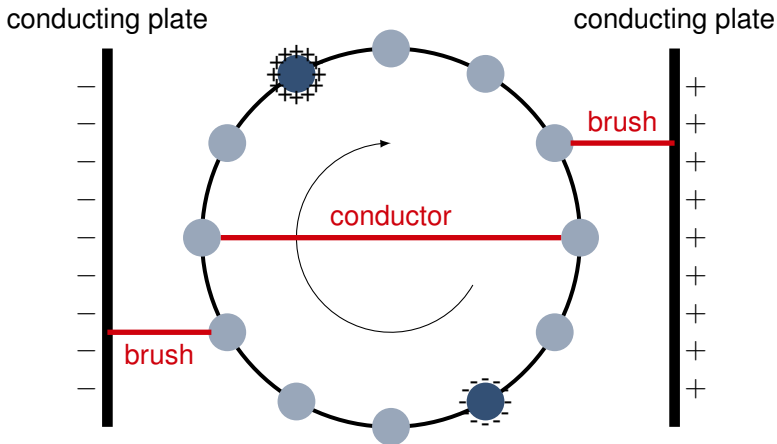
A dirod generator works by inducing charges on the rotating cylinders and transferring those charges to plates.

The dirod generator



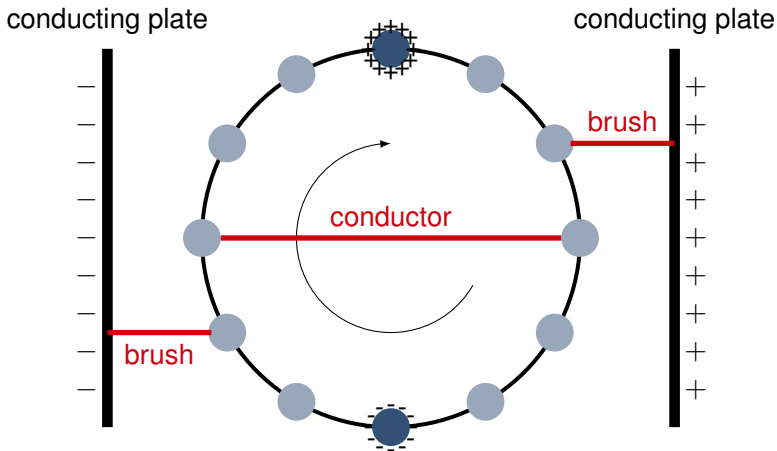
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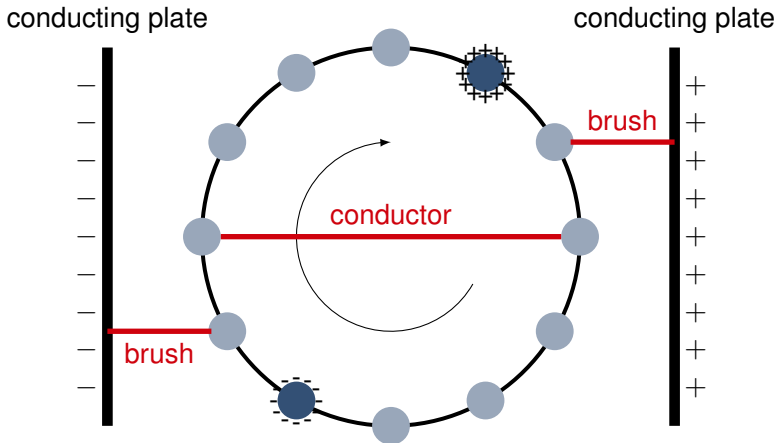
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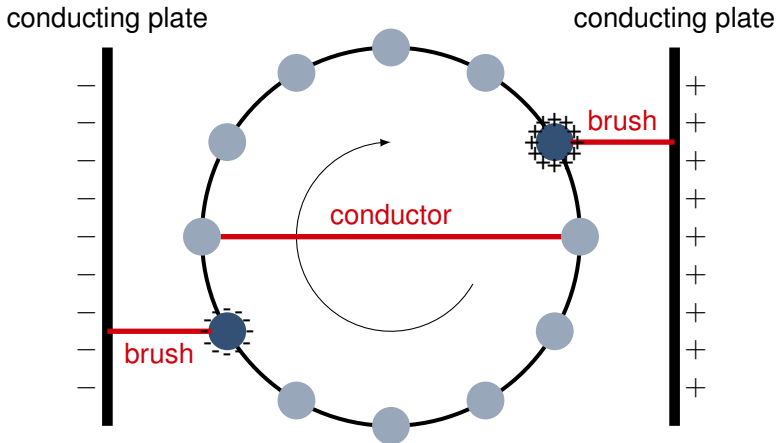
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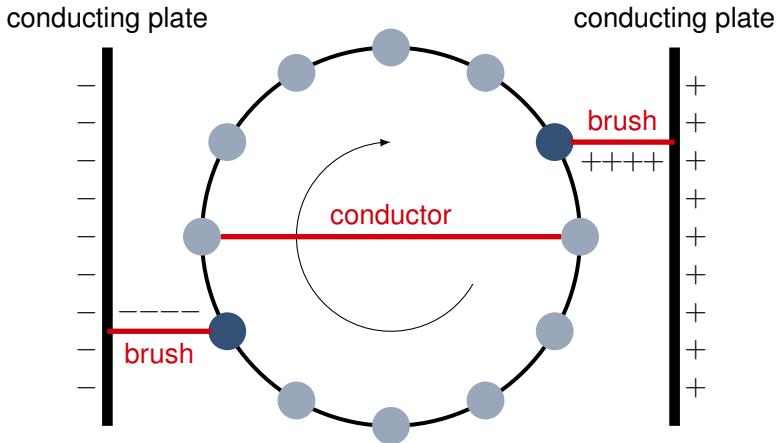
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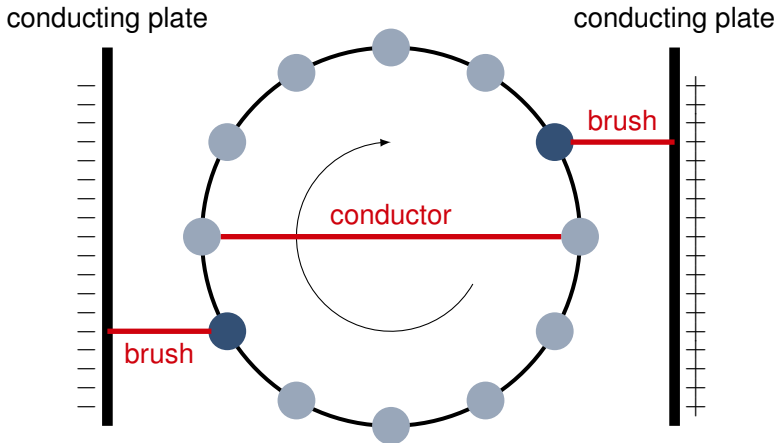
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Coloumb's Law

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Induction, Coulomb and The Electric Field

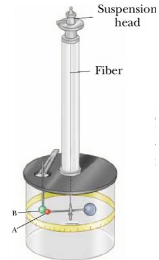
What did Coulomb observe?

- Force is proportional to charges and the direction depends on their relative sign:

$$F \propto Q_1 Q_2$$



https://en.wikipedia.org/wiki/Charles-Augustin_de_Coulomb



What did Coulomb observe?

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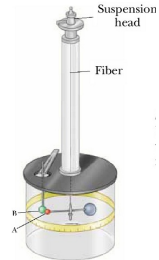
$$F \propto Q_1 Q_2$$

- Both charges feel equal and opposite force:

$$\vec{F}_{12} = -\vec{F}_{21}$$



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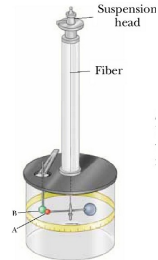
$$\vec{F}_{12} = -\vec{F}_{21}$$

- Inversely proportional to distance squared:

$$F \propto \frac{1}{r^2}$$



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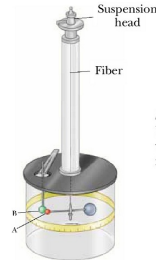
$$\vec{F}_{12} = -\vec{F}_{21}$$

- Inversely proportional to distance squared:

$$F \propto \frac{1}{r^2}$$
$$\therefore F \propto \frac{Q_1 Q_2}{r^2} = k \frac{Q_1 Q_2}{r^2}$$



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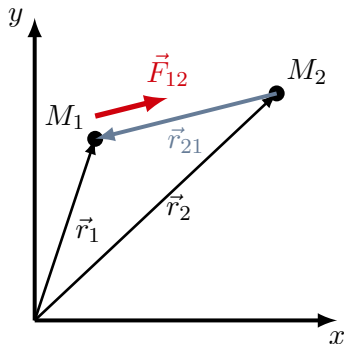
Recall Newton's Universal Theory of Gravity

The force, $\vec{F}_{12}$, **on** mass M_1 **from** M_2 can be written as:

$$\vec{F}_{12} = -G \frac{M_1 M_2}{r_{21}^2} \hat{r}_{21} = -G \frac{M_1 M_2}{r_{21}^3} \vec{r}_{21}$$

where:

- $\vec{r}_{21} = \vec{r}_2 - \vec{r}_1$ is the vector from M_2 to M_1 .
- $\hat{r}_{21}$ is the unit vector in the $\vec{r}_{21}$ direction.
- $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ is the Universal Gravitational Constant.
- The force is always attractive due to the minus sign at the front.



Vectors in Newton's Universal Theory of Gravity. The force on M_1 is in the opposite direction of $\vec{r}_{21}$.

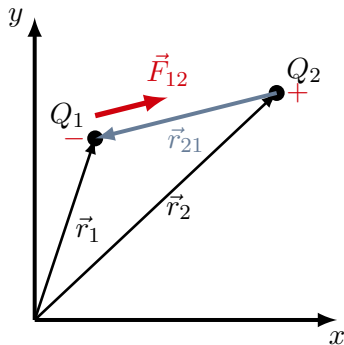
Coulomb's Law

The force, $\vec{F}_{12}$, **on** charge Q_1 **from** charge Q_2 can be written as:

$$\vec{F}_{12} = k \frac{Q_1 Q_2}{r_{21}^2} \hat{r}_{21} = k \frac{Q_1 Q_2}{r_{21}^3} \vec{r}_{21}$$

where:

- $\vec{r}_{21} = \vec{r}_2 - \vec{r}_1$ is the vector from Q_2 to Q_1 .
- $\hat{r}_{21}$ is the unit vector in the $\vec{r}_{21}$ direction.
- $k = 9.989 \times 10^9 \text{ Nm}^2/\text{C}^2$ is Coulomb's constant.
- The force can be **attractive or repulsive**, depending on the product $Q_1 Q_2$. If the charges are opposite, the product is negative and the force is attractive.



Vectors in Coulomb's Law. The force is attractive if the charges have opposite signs.

Gravity and the Electric force

- Mathematically, these are almost identical; instead of mass, we have electric charge (can think of gravitational mass as gravitational charge).
- The main difference is that electric charges can be positive or negative, which can make the force attractive or repulsive.
- Since we can define a gravitational field, $\vec{g}$, we should be able to define an “electric” field, $\vec{E}$:

$$\vec{F}_g = m\vec{g} \quad \vec{F}_E = q\vec{E}$$

- Since Gauss’ Law can allow us to determine the gravitational field, it should also work for the electric field (next week’s topic!).

Inverse square laws and waves

Recall: to conserve energy, the intensity of a wave must decrease with the square of distance.

- So, maybe, the “forces at a distance” (gravity and Coulomb) are somehow transmitted by a wave?
- Coulomb Force is actually transmitted by light waves!
- Gravity is transmitted by waves (Nobel Prize 2017)!

The Electric Field

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The electric field

- Consider the electric force from 2 stationary charges:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 = k \frac{qQ_1}{r_1^2} \hat{r}_1 + k \frac{qQ_2}{r_2^2} \hat{r}_2$$

- Since q is a scalar, it can be factored out:

$$\vec{F} = q \left(k \frac{Q_1}{r_1^2} \hat{r}_1 + k \frac{Q_2}{r_2^2} \hat{r}_2 \right)$$

- We can easily get the force vector for $q \rightarrow -2q$ by just multiplying the part in parenthesis by $-2q$ instead of q .
- The part in parenthesis is Force per Charge, or Electric field:

$$\vec{E} = \frac{\vec{F}}{q}$$

Recall, gravitational field outside a spherical mass

- At some distance r from a (spherical) object of mass M , a mass m feels a force:

$$\vec{F} = -G \frac{mM}{r^2} \hat{r}$$

- Force per unit mass (gravitational field) outside of a spherical body:

$$\vec{g} = \frac{F}{m} = -G \frac{M}{r^2}$$

- At the surface of the Earth (of radius, R):

$$\vec{g} = \frac{F}{m} = -G \frac{M}{R^2}$$

Electric field

- If you have a point charge, Q at the origin, you can calculate the electric field vector at any point in space, at a position $\vec{r}$:

$$\vec{E} = k \frac{Q}{r^2} \hat{r}$$

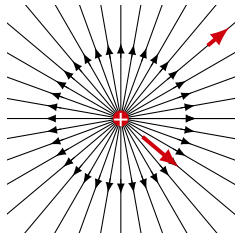
- To find the force vector on some other charge, q , you just multiply the value of the electric field by q :

$$\vec{F}_q = q\vec{E}$$

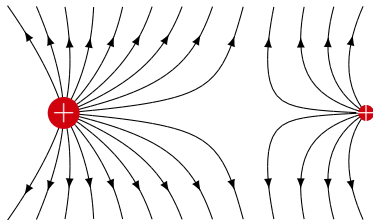
- The electric field is a “Vector Field” (it is a function that you can evaluate anywhere in space to obtain a vector).

Visualizing the electric field: Field Lines

- We can draw “field lines” that tell us how to determine the electric field vector at some point in space:
 - The more lines there are in an area of space, the stronger the electric field (the longer the vector) and corresponding force.
 - The electric field vector is always tangent to the field lines.
 - The electric field vector is always **in the direction of the force that a positive charge would feel**.



Electric field lines out of a positive point charge.

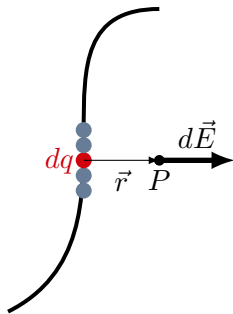


Electric field lines from two different positive charges.

Electric field and superposition

- If there are multiple charges, one can add the electric field from each charge to get the total electric field at one point in space.
- If we have continuous charge distributions (as opposed to point charges), then we can model those distributions as being the sum of many individual point charges.
- The total electric field can be found by adding the infinitesimal electric fields from each (point) charge element:

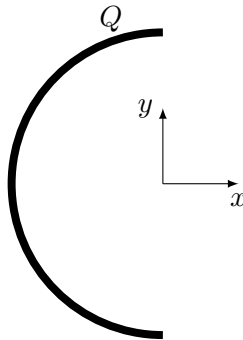
$$d\vec{E} = k \frac{dq}{r^2} \hat{r}$$
$$\vec{E} = \int d\vec{E}$$



To determine the electric field at point P from a charge distribution, model the distribution as a collection of point charges, dq , and add those fields together.

Electric field from a half-ring of charge

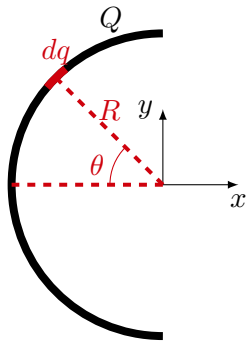
1. Make a *good* diagram.



Modeling the electric field from a half-ring of charge, Q .

Electric field from a half-ring of charge

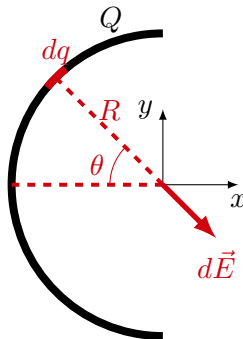
1. Make a *good* diagram.
2. Choose a charge element, dq .



Modeling the electric field from a half-ring of charge, Q .

Electric field from a half-ring of charge

1. Make a *good* diagram.
2. Choose a charge element, dq .
3. Draw the electric field element, $d\vec{E}$, due to dq .

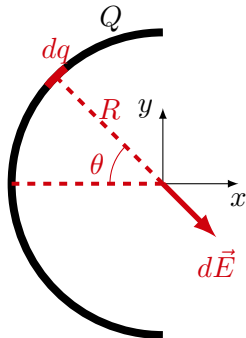


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1. Make a *good* diagram.
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4. Write the electric field element, $d\vec{E}$, using relevant variables:

$$d\vec{E} = \frac{k dq}{R^2} (\cos \theta \hat{x} - \sin \theta \hat{y}) = dE_x \hat{x} + dE_y \hat{y}$$



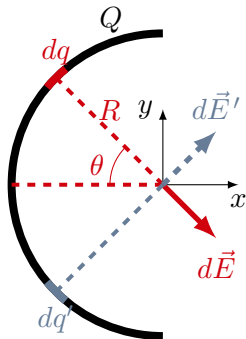
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5. Think of symmetry! ($\int dE_y = 0$)



Modeling the electric field from a half-ring of charge, Q .

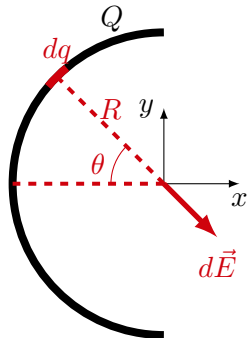
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6. Write the electric field as the sum of the $d\vec{E}$:

$$\vec{E} = \int d\vec{E} = \left(\int dE_x \right) \hat{x} + \left(\int dE_y \right) \hat{y}$$



Modeling the electric field from a half-ring of charge, Q .

Electric field from a half-ring of charge

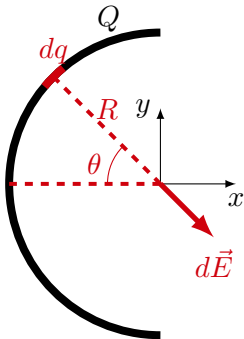
7. We will choose, θ , as the the integration variable:

$$E_x = \int dE_x = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{k dq}{R^2} \cos \theta$$

8. Introduce charge per unit length, $\lambda = \frac{Q}{\pi R}$:

$$dq = \lambda R d\theta = \frac{Q}{\pi R} R d\theta = \frac{Q}{\pi} d\theta$$

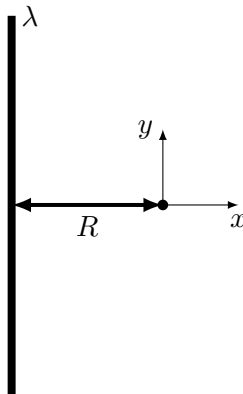
$$\begin{aligned} \therefore E_x &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{kQ}{\pi R^2} \cos \theta d\theta \\ &= \frac{kQ}{\pi R^2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta d\theta = \frac{kQ}{\pi R^2} [\sin \theta]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{2kQ}{\pi R^2} \end{aligned}$$



Modeling the electric field from a half-ring of charge, Q .

Electric field from an infinite line charge

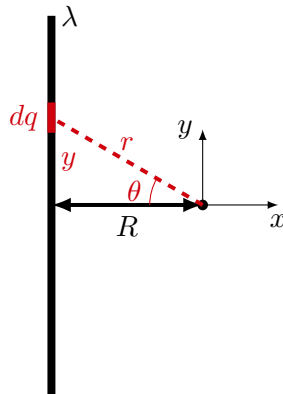
1. Make a *good* diagram.



Modeling the electric field at a distance, R , from an infinite line of charge.

Electric field from an infinite line charge

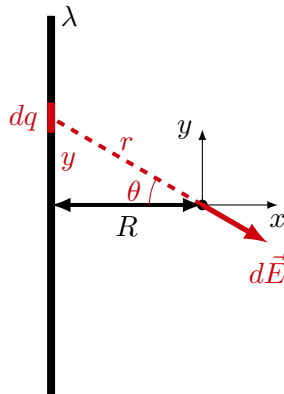
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2. Choose a charge element, dq .



Modeling the electric field at a distance, R , from an infinite line of charge.

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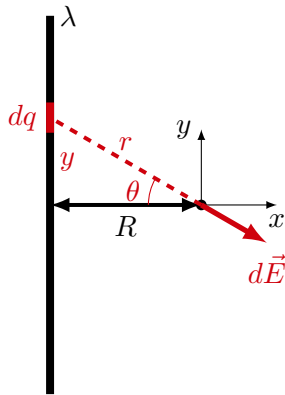


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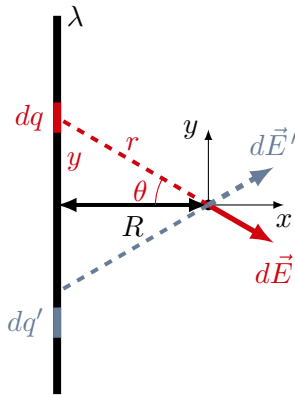
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5. Think of symmetry! ($\int dE_y = 0$):

$$\vec{E} = \left(\int \frac{k dq}{r^2} \cos \theta \right) \hat{x}$$



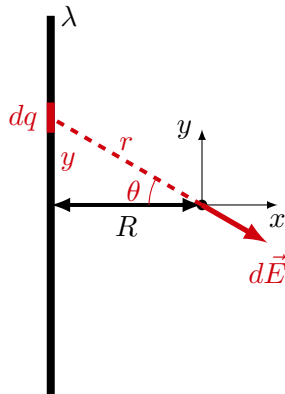
Modeling the electric field at a distance, R , from an infinite line of charge.

Electric field from an infinite line charge

$$\vec{E} = \left(\int \frac{k dq}{r^2} \cos \theta \right) \hat{x}$$

6. We can integrate over θ or y , we choose y .
Anything that changes with different dq needs
to be expressed with y and constants:

$$dq = \lambda dy \quad \text{and} \quad r^2 = R^2 + y^2 \quad \text{and} \quad \cos \theta = \frac{R}{\sqrt{R^2 + y^2}}$$



*Modeling the electric field at a distance, R ,
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Electric field from an infinite line charge

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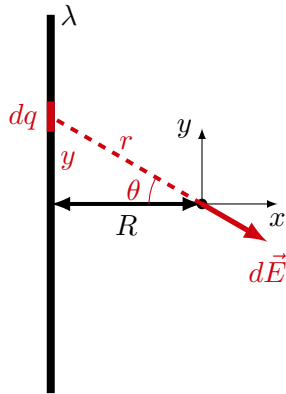
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7. All together:

$$E_x = \int_{-\infty}^{+\infty} \frac{k \lambda dy R}{(y^2 + R^2)^{\frac{3}{2}}} = k \lambda R \int_{-\infty}^{+\infty} \frac{dy}{(y^2 + R^2)^{\frac{3}{2}}}$$

ask your math friend for help, look it up...

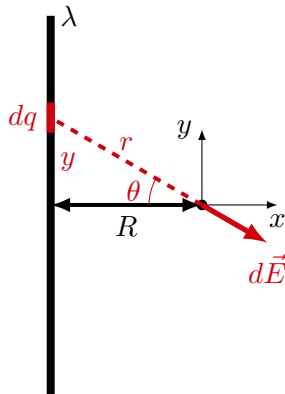


*Modeling the electric field at a distance, R ,
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Electric field from an infinite line charge

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- What if we integrate over θ ?



Modeling the electric field at a distance, R , from an infinite line of charge.

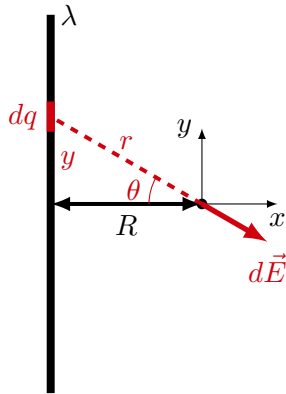
Electric field from an infinite line charge

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- What if we integrate over θ ?

$$y = R \tan \theta \rightarrow \frac{dy}{d\theta} = \frac{R}{\cos^2 \theta}$$

$$\therefore dq = \lambda dy = \lambda \frac{R}{\cos^2 \theta} d\theta \quad \text{and} \quad r^2 = \frac{R^2}{\cos^2 \theta}$$



Modeling the electric field at a distance, R , from an infinite line of charge.

Electric field from an infinite line charge

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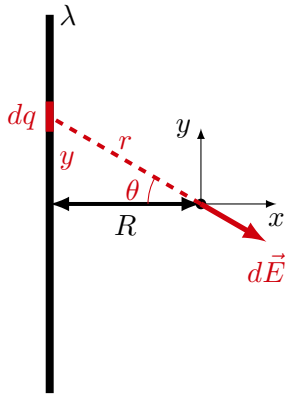
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$$\therefore dq = \lambda dy = \lambda \frac{R}{\cos^2 \theta} d\theta \quad \text{and} \quad r^2 = \frac{R^2}{\cos^2 \theta}$$

- All together:

$$\begin{aligned} E_x &= \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} k \lambda \frac{R d\theta}{\cos^2 \theta} \frac{\cos^2 \theta}{R^2} \cos \theta = \frac{k \lambda}{R} \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \cos \theta d\theta \\ &= \frac{2k\lambda}{R} \end{aligned}$$



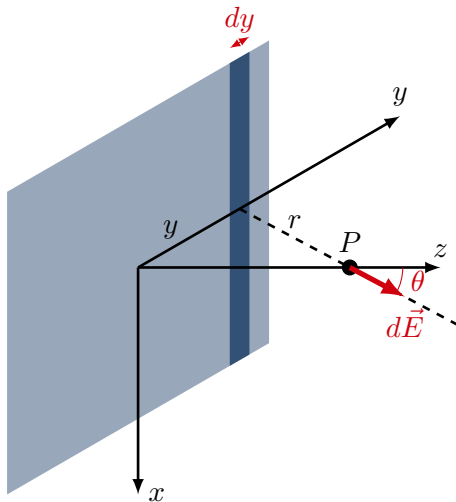
Modeling the electric field at a distance, R , from an infinite line of charge.

2D infinite plane

- If you have a continuous 2D (3D) charge distribution, model it as the sum of electric fields from 1D (2D) charge distributions.
- For example, for an infinite plane (2D), consider it has being made of many infinite lines of charge (1D), for which you just calculated the electric field.
 - The plane has surface charge density, σ .
 - An infinite line of charge with width dy thus has linear charge density:

$$\lambda = \sigma dy$$

- Sum all of the electric fields from lines of charge at different values of y .



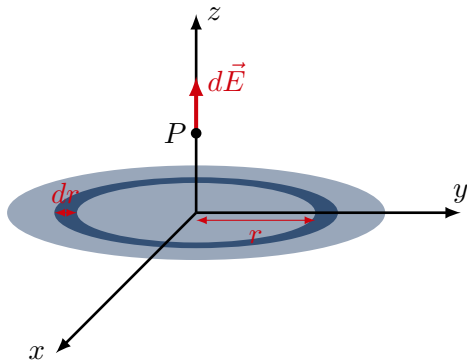
The electric field from an infinite plane with charge per unit area, σ . By symmetry, the net field will be in the y direction.

2D infinite plane, different method

- You can model the infinite plane as the sum of rings.
- A ring of width dr has total charge given by:

$$dq = \sigma 2\pi r dr$$

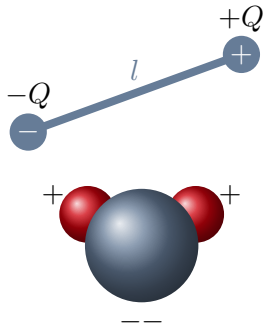
- Sum all of the electric fields from all of the rings, from $r = 0$ to $r = \infty$.



The electric field from an infinite plane with charge per unit area, σ . By symmetry, the net field will be in the y direction.

Electric dipoles

- Dipoles are a pair of equal magnitude but opposite sign charges, Q , separated by a fixed distance, l .
- For example, a water molecule can be modeled as a dipole.



A dipole and a water molecule.

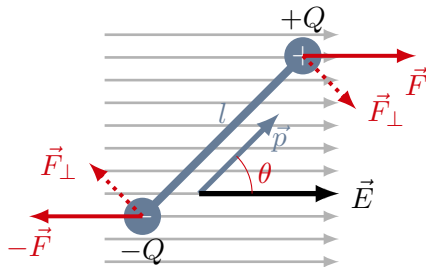
The dipole moment vector, $\vec{p}$

- We introduce the **dipole vector**, $\vec{p}$:
 - Magnitude: $p = Ql$
 - Direction from $-Q$ to $+Q$
- The net torque on the dipole is then given by:

$$\vec{\tau} = \vec{p} \times \vec{E}$$
$$\tau = pE \sin \theta$$

Check:

$$\begin{aligned} F_{\perp} &= QE \sin \theta \\ \therefore \tau &= F_{\perp} \frac{l}{2} + F_{\perp} \frac{l}{2} \\ &= QEl \sin \theta = pE \sin \theta \end{aligned}$$



A dipole, formed by two charges of magnitude, Q , a distance, l , apart in an electric field, $\vec{E}$.

Dipole potential energy

- The potential energy is minimum when the dipole vector is parallel to the electric field.
- The potential energy is maximum when the dipole vector is anti-parallel to the electric field.
- We can write the potential energy:

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

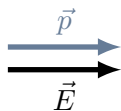


Diagram showing a dipole vector $\vec{p}$ (blue arrow) parallel to an electric field vector $\vec{E}$ (black arrow). The potential energy is $U = -pE$.

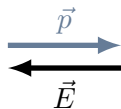


Diagram showing a dipole vector $\vec{p}$ (blue arrow) anti-parallel to an electric field vector $\vec{E}$ (black arrow). The potential energy is $U = pE$.

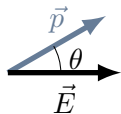


Diagram showing a dipole vector $\vec{p}$ (blue arrow) at an angle θ to an electric field vector $\vec{E}$ (black arrow). The potential energy is $U = -pE \cos \theta$.

A dipole in an electric field.